

Introduction to Electronics Engineering

MODULE 3 — PART 1: COMPLETE NOTES

Module 3 (07 hrs): Sinusoidal Oscillators & Operational Amplifiers

For First-Year Students — written in simple, easy language
Course: Introduction to Electronics Engineering (IEE) · Course Code: EC 101

Module 3 — what it covers (as per the syllabus)

- Sinusoidal Oscillators: concept of positive and negative feedback
- Barkhausen criterion, determination of frequency and condition of oscillation
- RC phase-shift and Wien bridge oscillators
- Operational Amplifiers: characteristics, applications of inverting and non-inverting amplifiers

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1	Feedback Amplifiers (positive and negative feedback)
2	Oscillators and the Barkhausen Criterion
3	Operational Amplifiers (characteristics, equivalent circuit, ideal vs practical)
4	Inverting Amplifier
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6	Summing Amplifier
7	RC Phase-Shift Oscillator
8	Wien Bridge Oscillator
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1. Feedback Amplifiers

1.1 What is Feedback?

The purpose of an amplifier is to amplify the input signal without changing its characteristics except its amplitude. An amplifier that works on the principle of feedback is called a **feedback amplifier**.

Feedback is a process in which a fraction of the output (voltage or current) is fed back to the input. A simple block diagram of a feedback amplifier is shown below.

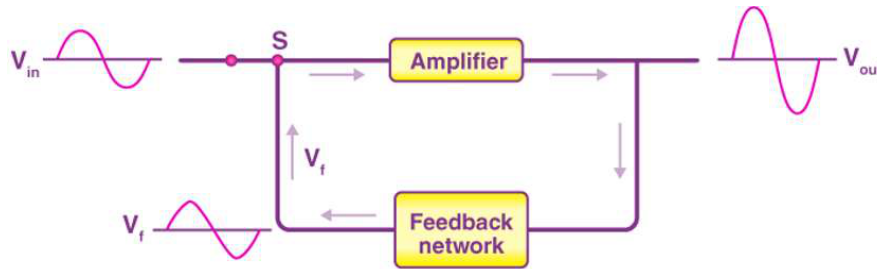


Figure 1: Block diagram of a feedback amplifier.

Block diagram of a feedback amplifier (positive / negative feedback)

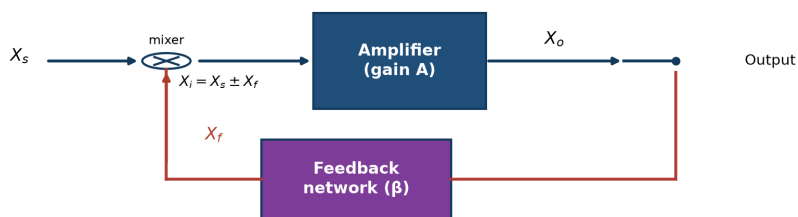


Figure 2: A feedback amplifier — the input is combined with a feedback signal in a mixer, amplified, and a part of the output is returned through the feedback network.

1.2 Working of the Feedback Amplifier

- The input signal V_s is applied to a **mixer network**, in which it is combined with a feedback signal V_f .
- The difference (or sum) of these signals, V_i , is the input to the amplifier.
- A portion of the amplifier output is connected back to the input through a **feedback network**.

$$V_i = V_s \pm V_f$$

- A = Open-loop gain
- β = Feedback factor
- V_s = Source voltage
- V_i = Input to amplifier
- V_o = Output voltage

Figure 3: The feedback signal is combined with the input at the mixer.

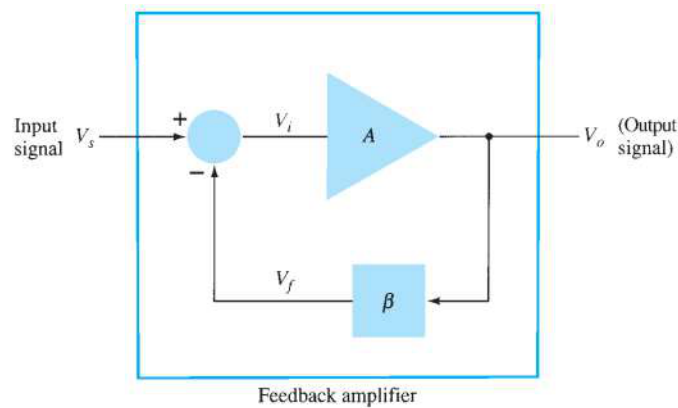


Figure 4: Gain relations of the feedback amplifier.

1.3 Positive Feedback

Positive feedback is a process in which a portion of the output signal is fed back to the input **in the same phase**, thereby increasing the net input signal.

- There is an in-phase relationship between input and output.
- It increases the net gain of the system.
- It is used in oscillators.

1.4 Negative Feedback

Negative feedback is a process in which a portion of the output signal is fed back to the input **in opposite phase**, thereby reducing the net input signal.

- There is an out-of-phase relationship between input and output.
- Its gain value is less than that of positive feedback.
- Its stability is higher than that of positive feedback.

Closed-loop gain (negative feedback): $A_f = A / (1 + \beta A)$ **(positive feedback: $A_f = A / (1 - \beta A)$)**

1.5 Positive vs. Negative Feedback

Parameter	Negative Feedback	Positive Feedback
Phase relationship	Out of phase (opposite phase)	In phase
Gain	Reduced	Increased
Stability	Increases	Decreases
Distortion	Reduces	Increases
Used in	Amplifiers	Oscillators

2. Oscillators and the Barkhausen Criterion

2.1 What is an Oscillator?

- The use of positive feedback that results in a feedback amplifier having a closed-loop gain greater than 1 and satisfying the phase conditions will result in operation as an **oscillator circuit**.
- An oscillator circuit is capable of producing an AC voltage of the desired frequency and waveshape.
- If the output signal varies sinusoidally, the circuit is called a **sinusoidal oscillator**.
- If the output voltage rises quickly to one voltage level and later drops quickly to another level, the circuit is called a **pulse or square-wave oscillator**.

2.2 Barkhausen Criterion for Oscillation

An amplifier with positive feedback oscillates when the loop gain is exactly unity: this is the **Barkhausen criterion**.

$$\text{beta } A = 1$$

- The output waveform will still exist after the switch is closed if the condition $\text{beta } A = 1$ is met.
- In reality, no input signal is needed to start the oscillator. Only the condition $\text{beta } A = 1$ must be satisfied for self-sustained oscillations.
- In practice, with $\text{beta } A > 1$ the system is started oscillating by amplifying the noise voltage; the resulting waveforms are never exactly sinusoidal.
- The closer the value of $\text{beta } A$ is to exactly 1, the more nearly sinusoidal is the waveform.

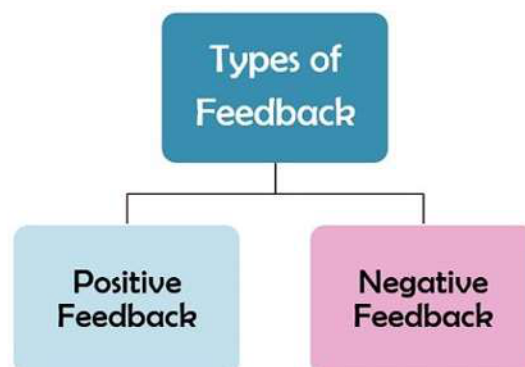


Figure 5: Conditions of oscillation for different values of the loop gain $\text{beta } A$.

2.3 Oscillator Operation — Start-up and Sustained Oscillation

Depending on the value of the loop gain $\text{beta } A$, the output behaves in one of three ways:

- **$\text{beta } A < 1$** — the oscillations are damped and gradually die out.
- **$\text{beta } A = 1$** — the oscillations are sustained at a constant amplitude (ideal oscillation).
- **$\text{beta } A > 1$** — the oscillations grow in amplitude until some limiting action brings them to a steady state.

- $|A\beta| < 1 \rightarrow$ Oscillations die out
- $|A\beta| > 1 \rightarrow$ Amplitude grows (distortion)
- $|A\beta| = 1 \rightarrow$ Sustained oscillations

Figure 6: Decaying (damped) oscillation when the loop gain is less than 1.

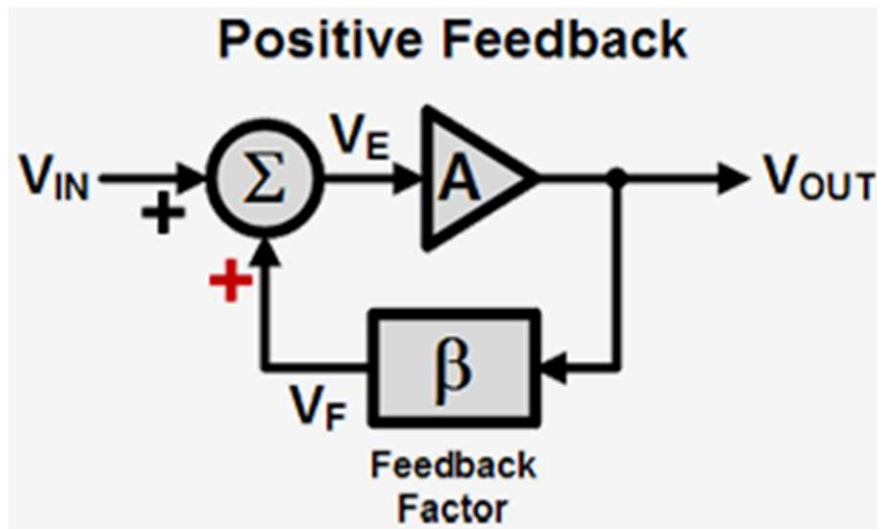


Figure 7: Sustained oscillation when the loop gain equals 1.

1. Magnitude Condition

$$|A\beta| = 1$$

2. Phase Condition

$$\angle A\beta = 0^\circ \text{ or } 360^\circ$$

Figure 8: Growing oscillation when the loop gain is greater than 1.

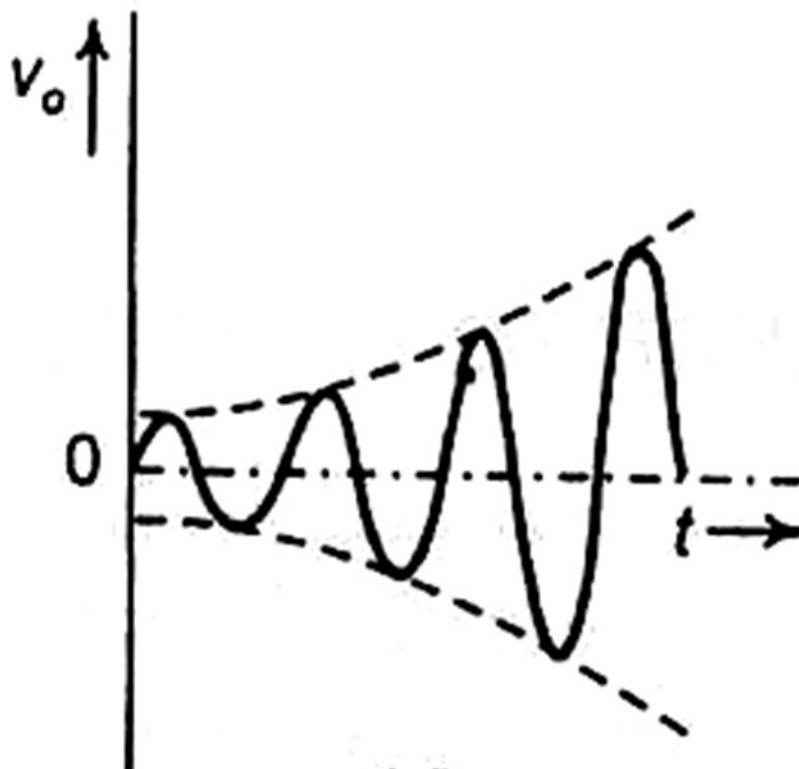


Figure 9: Oscillator output waveform (start-up to steady state).

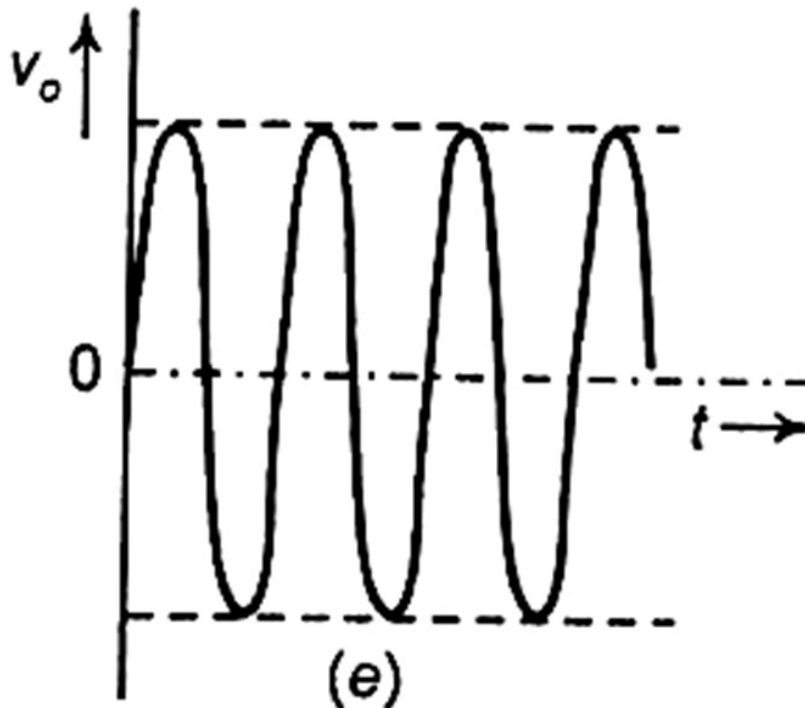


Figure 10: Oscillator output waveform — growth of amplitude.

3. Operational Amplifiers (Op-Amps)

3.1 Introduction and Symbol

- An op-amp is so named because it was originally designed to perform mathematical operations such as addition, subtraction, multiplication, division, integration and differentiation in analog computers.
- An **Operational Amplifier (Op-Amp)** is a high-gain voltage amplifier with two inputs and one output. It amplifies the **difference** between the two input voltages.
- The op-amp is represented by a triangular symbol with two input terminals and one output terminal.
- The terminal with the negative sign is the **inverting input**; the terminal with the positive sign is the **non-inverting input**.

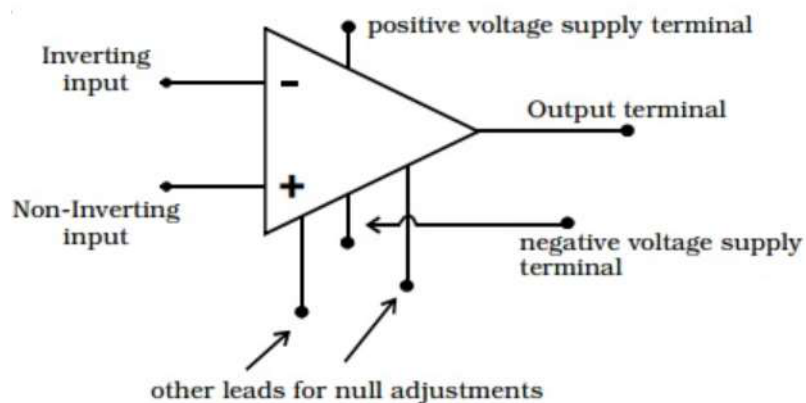


Figure 11: Symbol of the operational amplifier — inverting (–) and non-inverting (+) inputs, one output.

$$V_o = A_{OL} V_d = A_{OL} (V_1 - V_2)$$

The output voltage is directly proportional to the difference voltage V_d . The common practical op-amp is the 741 ($\mu A741$), whose pin diagram is shown below.

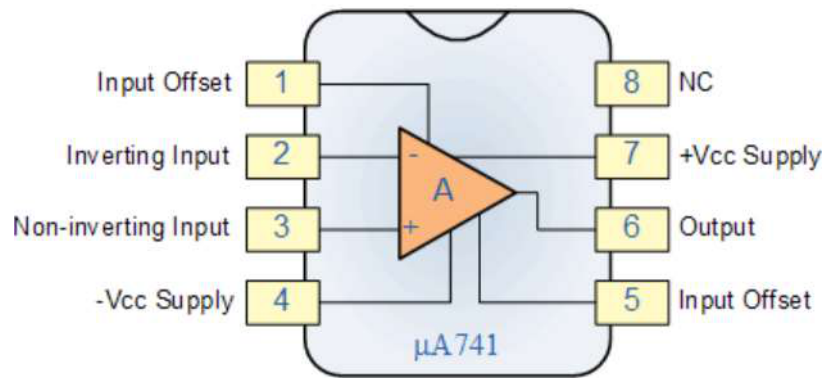


Figure 12: Pin diagram of the $\mu A741$ op-amp.

3.2 Equivalent Circuit of the Op-Amp

The circuit which represents the op-amp parameters in terms of physical components, for analysis purposes, is called the **equivalent circuit** of the op-amp. It shows the input resistance R_{in} , output resistance R_o and the open-loop voltage gain in terms of circuit components.

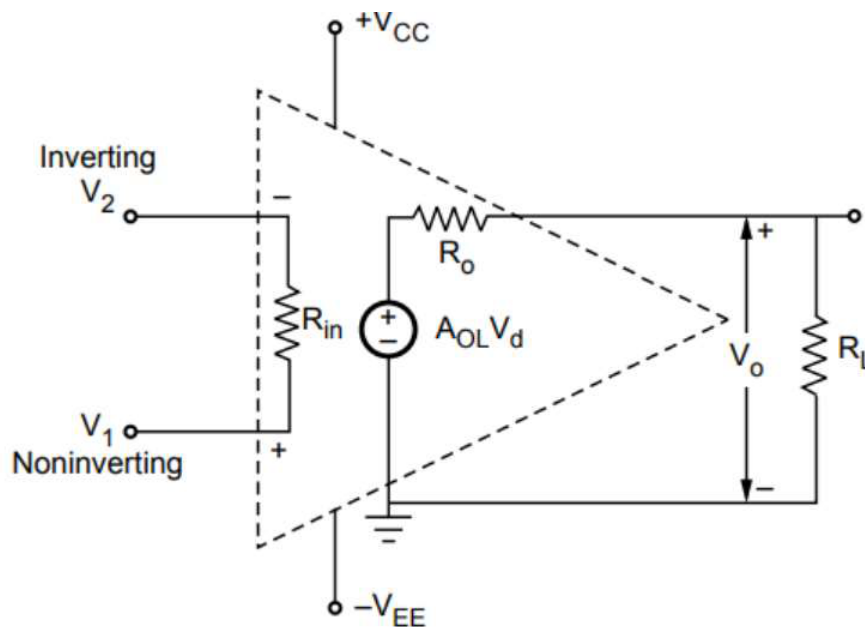


Figure 13: Equivalent circuit of the op-amp — R_{in} , the controlled source $A_{OL} V_d$ and R_o .

$$V_o = A_{OL} V_d = A_{OL}(V_1 - V_2)$$

where A_{OL} = Large signal open loop voltage gain.

V_d = Difference voltage $V_1 - V_2$.

V_1 = Noninverting input voltage with respect to ground.

V_2 = Inverting input voltage with respect to ground.

R_i = Input resistance of op-amp.

R_o = Output resistance of op-amp.

Figure 14: Meaning of each parameter used in the equivalent circuit.

- The op-amp amplifies the **difference voltage**, not the individual input voltages. Thus the output polarity is decided by the polarity of the difference voltage V_d .
- The voltage source $A_{OL} V_d$ is the Thevenin equivalent voltage source, while R_o is the Thevenin equivalent resistance looking back into the output terminals.

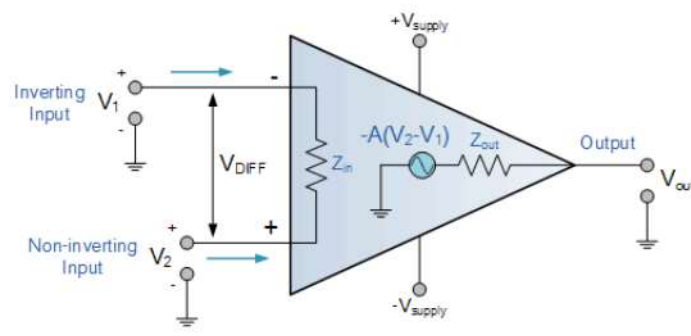


Figure 15: Equivalent circuit showing the input impedance Z_{in} , output impedance Z_{out} and the controlled source.

3.3 Characteristics of an Ideal Op-Amp

a) Infinite voltage gain ($A = \text{infinity}$)

A is the differential open-loop gain, and it is infinite for an ideal op-amp.

b) Infinite input impedance ($Z_{in} = \text{infinity}$)

This ensures that no current can flow into an ideal op-amp. An ideal op-amp draws no current at either input terminal, i.e. $I_1 = I_2 = 0$, so its input impedance is infinite.

c) Zero output impedance ($Z_{out} = 0$)

This ensures that the output voltage of the op-amp remains the same irrespective of the value of the load resistance connected.

d) Zero offset voltage ($V_{ios} = 0$)

The presence of a small output voltage even when $V_1 = V_2 = 0$ is called an offset voltage. It is zero for an ideal op-amp, which ensures zero output for a zero input signal.

e) Infinite bandwidth

The range of frequency over which the amplifier performance is satisfactory is called its bandwidth. The bandwidth of an ideal op-amp is infinite.

f) Infinite slew rate (S = infinity)

This ensures that changes in the output voltage occur simultaneously with changes in the input voltage.

Slew rate: the maximum rate of change of output voltage per unit time, expressed in volts per microsecond (V/μs).

$$SR = \frac{\Delta V_o}{\Delta t} \text{ V}/\mu\text{s} \quad \text{with } t \text{ in } \mu\text{s}$$

Figure 16: Definition of slew rate.

g) Infinite CMRR (CMRR = infinity)

The common-mode rejection ratio (CMRR) is a measure of the amplifier's ability to reject unwanted common-mode signals such as noise. It is defined as the ratio of the differential gain (A_d) to the common-mode gain (A_{cm}).

- **Common-mode gain A_{cm}** — the gain of the amplifier when the same signal is applied to both inputs.
- **Differential gain A_d** — the gain of the amplifier when it amplifies the difference between two input voltages.

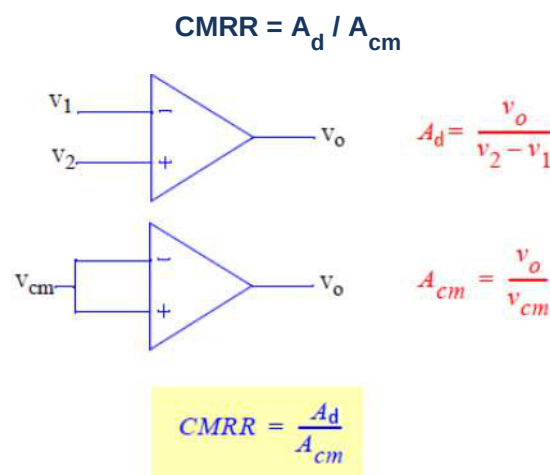


Figure 17: Differential gain and common-mode gain relations.

$$CMRR_{dB} = 20 \log_{10} \left(\frac{A_d}{A_{cm}} \right)$$

Figure 18: CMRR equation.

3.4 Ideal vs. Practical Op-Amp

Property	Ideal	Practical(Typical)
Open-loop gain	Infinite	Very high (>10000)
Open-loop bandwidth	Infinite	Dominant pole(≈ 10 Hz)
CMRR	Infinite	High (> 60 dB)
Input Resistance	Infinite	High (>1 M Ω)
Output Resistance	Zero	Low(< 100 Ω)
Input Bias Currents	Zero	Low (< 50 nA)
Offset Voltages	Zero	Low (< 10 mV)
Offset Currents	Zero	Low (< 50 nA)
Slew Rate	Infinite	A few V/ μ s

Figure 19: Comparison of the ideal and practical op-amp parameters.

Parameter	Ideal Op-Amp	Practical Op-Amp (741)
Open-loop gain (A)	Infinite	Very high (about 10^5)
Input impedance (Z_{in})	Infinite	High (about 2 M-ohm)
Output impedance (Z_{out})	Zero	Low (about 75 ohm)
Bandwidth	Infinite	Limited (about 1 MHz)
Offset voltage	Zero	Small but non-zero (mV)
Slew rate	Infinite	Limited (about 0.5 V/ μ s for 741)
CMRR	Infinite	High but finite (about 90 dB)

3.5 Applications of Op-Amps

Op-amps are used in a wide variety of applications, including:

- **Signal amplification** — to increase the amplitude of weak signals.
- **Filters** — in active filters (low-pass, high-pass, band-pass, etc.) for audio and communication applications.
- **Mathematical operations** — circuits that perform integration, differentiation, addition and subtraction.
- **Voltage follower (buffer)** — provides high input impedance and low output impedance to isolate stages without affecting the signal.
- **Oscillators and signal generators** — to generate waveforms like sine, square and triangular waves.
- **Analog computation** — in analog computers, to perform arithmetic and calculus functions.

4. Inverting Amplifier

4.1 Circuit and Virtual Ground

An **inverting operational amplifier** is a fixed-gain voltage amplifier configuration that amplifies an input signal while **inverting** its output by 180° compared to the input.

- An input signal V_1 is applied through resistor R_1 to the minus (inverting) input.
- The output is connected back to the same minus input through resistor R_f .
- The plus (non-inverting) input is connected to ground.
- Since the signal is essentially applied to the minus input, the resulting output is opposite in phase to the input.

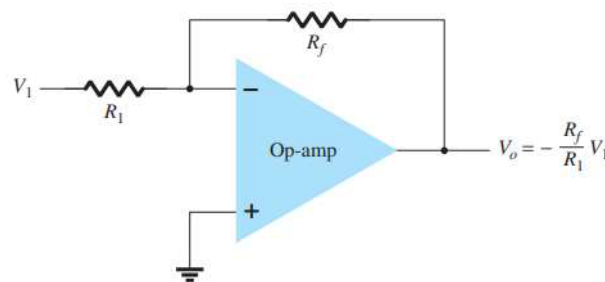


Figure 20: The inverting amplifier circuit.

Virtual ground: the inverting terminal is not physically connected to ground. It behaves like ground (0 V), but it is not actually connected to ground — this is called a **virtual ground**. The concept of a **virtual short** implies that although the voltage is nearly 0 V, there is no current through the amplifier input to ground.

4.2 Analysis and Gain

- No current enters the op-amp input (ideal op-amp).
- Applying KCL at the inverting node: the current through the input resistor equals the current through the feedback resistor.
- The inverting node is at virtual ground (0 V).
- The output is 180° phase-shifted with respect to the input.

$$I_1 = (V_1 - 0) / R_1 \quad \text{and} \quad I_f = (0 - V_o) / R_f$$

$$I_1 = I_f \Rightarrow V_1 / R_1 = -V_o / R_f$$

$$\text{Gain: } A_v = V_o / V_1 = -R_f / R_1$$

4.3 Characteristics and Applications

- **Phase inversion:** the output signal is 180° out of phase with the input.
- **Gain:** determined by the ratio of the feedback resistor to the input resistor, $A_v = -R_f / R_1$.
- **High input impedance (with feedback):** feedback helps achieve higher input impedance than the input resistor alone.
- **Virtual ground:** the inverting input is at the same potential as ground (0 V), but no current flows through it.

Applications:

- 1 **Summing amplifier** — can add multiple input signals, each with its own resistor.
- 2 **Analog integration and differentiation** — used in analog signal processing.
- 3 **Active filters** — used in active filter designs due to their controllable gain.

4 **Oscillators** — used to produce square waves or other periodic signals.

5. Non-Inverting Amplifier

5.1 Circuit and Working

A **non-inverting amplifier** is an op-amp configuration in which the input is applied to the non-inverting (plus) terminal, and the output is fed back to the inverting (minus) terminal through a resistor.

- The input signal is applied to the plus terminal.
- The output is fed back to the minus terminal through resistor R_f , with resistor R_1 connected to ground.
- Due to the virtual short, the voltages at the plus and minus terminals are nearly equal, and no current flows into the op-amp input.
- The current through R_1 equals the current through R_f .
- The output is in phase with the input.

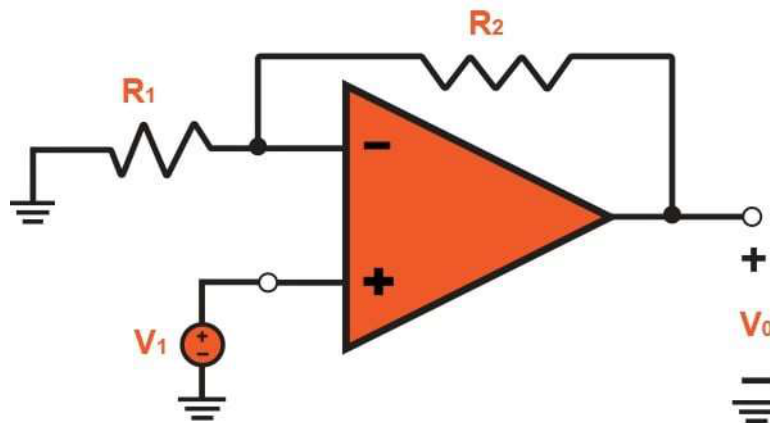


Figure 21: The non-inverting amplifier circuit.

5.2 Analysis and Gain

Because $V_+ = V_- = V_{in}$ (virtual short), the current through R_1 is V_{in}/R_1 and equals the current through R_f :

$$V_{in} / R_1 = (V_o - V_{in}) / R_f$$

$$\text{Gain: } A_{CL} = V_o / V_{in} = 1 + R_f / R_1$$

5.3 Characteristics and Applications

- **No phase inversion:** the output signal is in phase with the input.
- **Gain:** positive and given by $A_{CL} = 1 + R_f/R_1$.
- **High input impedance:** extremely high, often in the megohms or higher range.
- **Excellent signal fidelity:** accurate amplification without inversion.

Applications:

- 1 **Voltage follower (buffer):** when $R_f = 0$ and R_1 is infinite, the gain becomes 1 and the circuit acts as a buffer, offering high input and low output impedance without altering the signal.
- 2 **Precision amplifiers:** for precise, non-inverted amplification, such as sensor signal amplification.
- 3 **Audio preamplifiers:** to amplify signals without altering their phase, ensuring accurate sound reproduction.

- 4 **Data acquisition:** amplifies weak sensor signals without phase distortion, which is essential for accurate measurements.

5.4 Voltage Follower (Buffer)

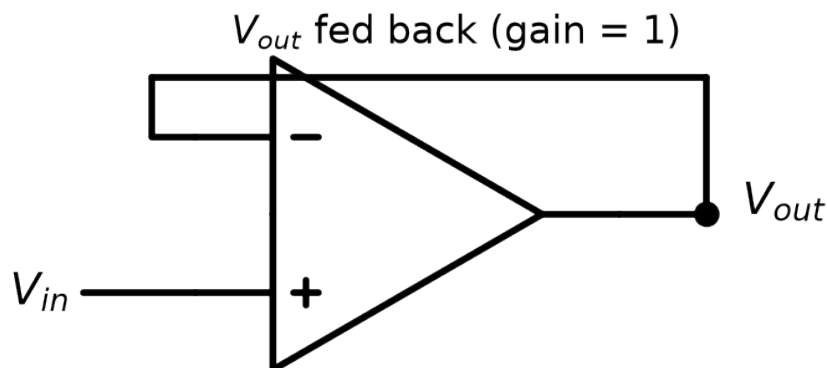


Figure 22: The voltage follower — the output is fed directly back to the inverting input, giving a gain of exactly 1.

5.5 Inverting vs. Non-Inverting Amplifier

Feature	Inverting Amplifier	Non-Inverting Amplifier
Input terminal	Inverting (–)	Non-inverting (+)
Output phase	180° phase shift (inverted)	In phase with input
Gain	$-R_f / R_1$	$1 + R_f / R_1$
Input impedance	Moderate	High
Applications	Summing, integration, filters, oscillators	Voltage follower, precision amplification

6. Summing Amplifier

The summing amplifier is a very useful op-amp circuit, built on the inverting amplifier. The circuit below shows a three-input summing amplifier, which provides a means of algebraically summing (adding) three voltages, each multiplied by a constant gain.

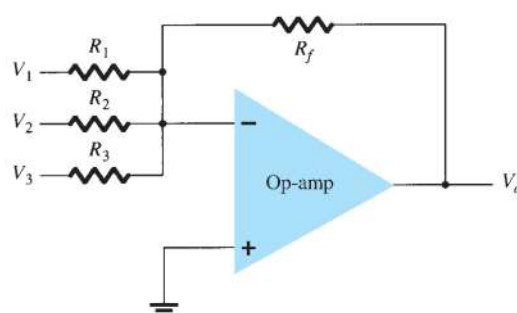


Figure 23: Three-input summing amplifier.

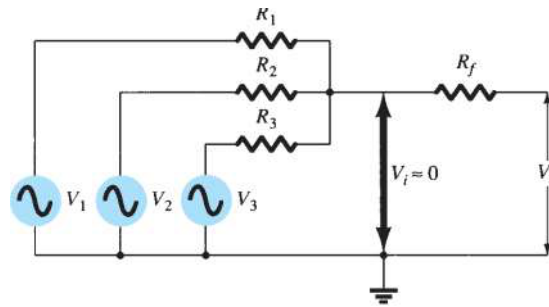


Figure 24: Equivalent circuit used for the analysis.

Applying KCL at the inverting node (which is at virtual ground):

$$V_1/R_1 + V_2/R_2 + V_3/R_3 = -V_o/R_f$$

Multiplying both sides by R_f , the output voltage is: $V_o = -R_f (V_1/R_1 + V_2/R_2 + V_3/R_3)$

If all resistors are equal ($R_1 = R_2 = R_3 = R_f = R$), then $V_o = -(V_1 + V_2 + V_3)$ — a simple adder.

7. RC Phase-Shift Oscillator

A **phase-shift oscillator** is a sinusoidal oscillator that produces a stable sine wave using an amplifier (op-amp or transistor) and an RC phase-shift network. The RC network provides the required phase shift so that the feedback becomes positive feedback, which is necessary for oscillation.

- For oscillation, βA must be > 1 , and the phase shift around the feedback network must be 180° (providing positive feedback).
- For the loop gain βA to be greater than unity, the gain of the amplifier stage must be greater than $1/\beta$, i.e. greater than 29.
- The values of R and C are chosen to provide (at a specific frequency) 60° phase shift per section for three sections, resulting in a total 180° phase shift, as desired.

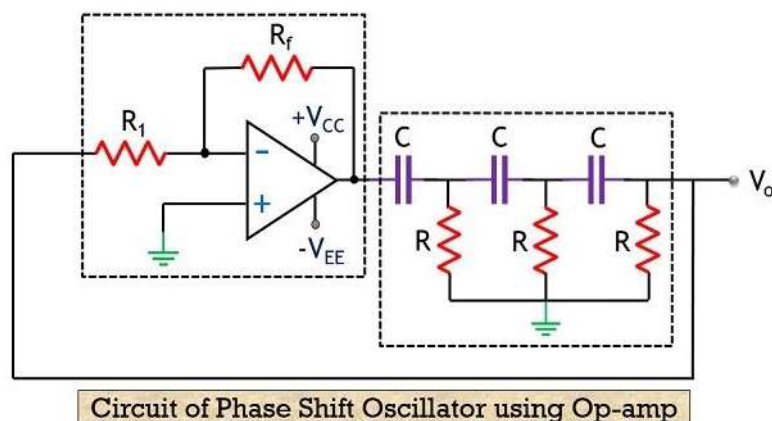


Figure 25: RC phase-shift oscillator using an op-amp.

- The output of the op-amp is fed to a three-stage RC network, which provides the needed 180° of phase shift at an attenuation factor of $1/29$.
- If the op-amp provides a gain (set by resistors R_1 and R_f) greater than 29, a loop gain greater than unity results and the circuit acts as an oscillator.

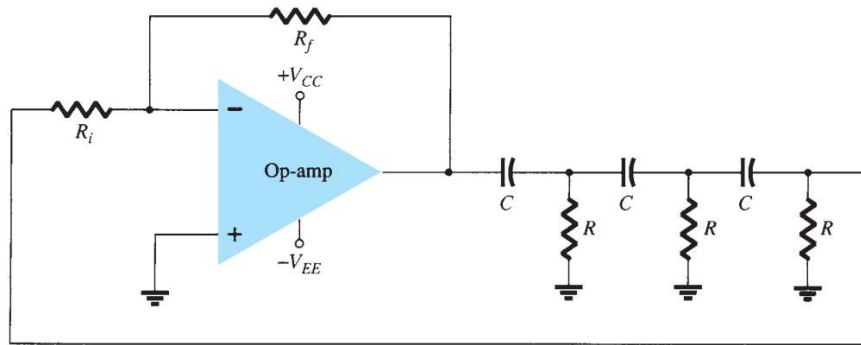


Figure 26: The three-section RC network and the op-amp amplifier.

Frequency of oscillation: $f = 1 / (2\pi RC\sqrt{6})$

Feedback factor: $\beta = 1/29$ Minimum amplifier gain required: $A > 29$

8. Wien Bridge Oscillator

A **Wien bridge oscillator** is a sinusoidal oscillator that uses an RC bridge network and an operational amplifier to generate low-distortion sine waves. It is widely used in audio frequency generators (20 Hz – 1 MHz).

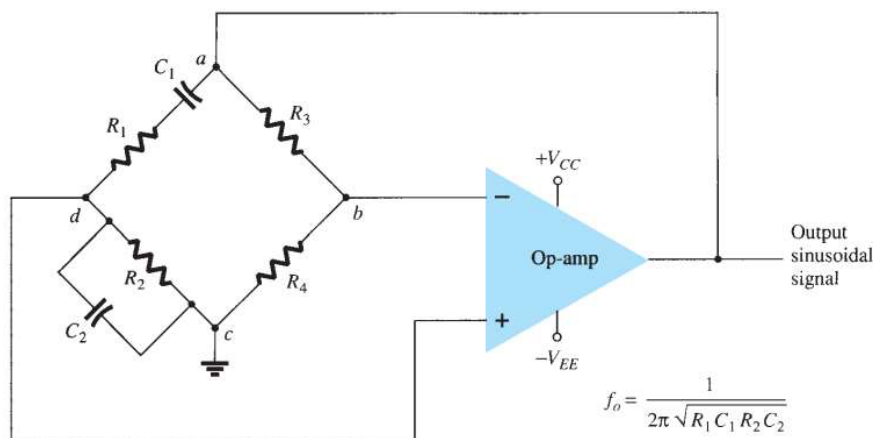


Figure 27: The Wien bridge oscillator circuit.

- Resistors R_1 and R_2 and capacitors C_1 and C_2 form the frequency-adjustment elements, while resistors R_3 and R_4 form part of the feedback path.
- The op-amp output is connected as the bridge input at points a and c.
- The bridge circuit output at points b and d is the input to the op-amp.

In a Wien bridge oscillator, the **resonant frequency** (oscillation frequency) is the frequency at which the Wien bridge RC network becomes balanced and produces **zero phase shift** between the input and output signals.

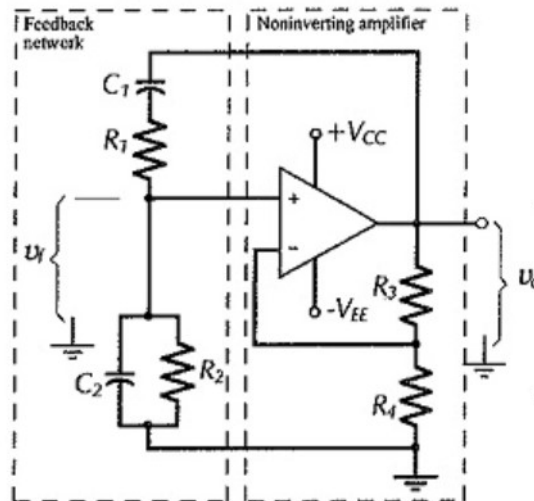


Figure 28: The Wien bridge network with the op-amp feedback path.

8.1 Frequency and Condition of Oscillation

At the resonant frequency:

- Phase shift = 0°
- Feedback fraction = $1/3$

Using the Barkhausen criterion, if $R_1 = R_2 = R$ and $C_1 = C_2 = C$, the resulting oscillator frequency is:

$$f_0 = 1 / (2 \pi R C)$$

Therefore the required amplifier gain is: $A = 3$, i.e. $R_3 / R_4 = 2$

Thus a ratio of R_3 to R_4 greater than 2 will provide sufficient loop gain for the circuit to oscillate at the frequency f_0 .

8.2 Advantages and Applications

Advantages:

- 1 The overall gain of the oscillator is high, as it uses a two-stage amplifier.
- 2 As no inductors are used in the circuit, there is no issue of interference from external magnetic fields.
- 3 This oscillator produces a stable sine wave without any distortion.
- 4 The frequency of oscillation can be changed by changing the values of the capacitors or by using a variable resistor in the circuit.
- 5 The Wien bridge oscillator has good frequency stability.

Applications:

- Audio signal generators
- Function generators
- Electronic test equipment
- Communication systems
- Instrumentation circuits

9. Solved Numerical Example

Example 1: Wien Bridge Oscillator Frequency

Problem: the RC network of a Wien bridge oscillator consists of resistors and capacitors of values $R_1 = R_2 = 22 \text{ k-ohm}$ and $C_1 = C_2 = 100 \text{ pF}$. Calculate the frequency of oscillation.

Example 12.4 The RC network of a Wien bridge oscillator consists of resistors and capacitors of values $R_1 = R_2 = 22 \text{ k}\Omega$ and $C_1 = C_2 = 100 \text{ pF}$. Calculate the frequency of oscillation.

Figure 29: The given data of the problem.

Solution:

Here, $R = 22 \text{ k-ohm} = 22 \times 10^3 \text{ ohm}$ and $C = 100 \text{ pF} = 10^{-10} \text{ F}$

For a Wien bridge oscillator: $f_0 = 1 / (2 \pi R C)$

$$f_0 = 1 / (2 \pi \times 22 \times 10^3 \times 10^{-10}) = 72.34 \text{ kHz}$$

$$f_0 = \frac{1}{2\pi RC} = \frac{1}{2\pi \times 22 \times 10^3 \times 10^{-10}} = 72.34 \text{ kHz}$$

Figure 30: Substitution and final result.

10. Quick Reference — Key Formulae and Definitions

10.1 Key Formulae

Quantity	Formula	Notes
Feedback input	$V_i = V_s \pm V_f$	+ for positive, – for negative feedback
Closed-loop gain (negative FB)	$A_f = A / (1 + \text{beta } A)$	Gain reduces
Closed-loop gain (positive FB)	$A_f = A / (1 - \text{beta } A)$	Gain increases
Barkhausen criterion	$\text{beta } A = 1$	Condition for sustained oscillation
Op-amp output	$V_o = A_{OL}(V_1 - V_2)$	Amplifies the difference
CMRR	$\text{CMRR} = A_d / A_{cm}$	Rejection of common-mode noise
Inverting amplifier gain	$A_v = -R_f / R_1$	Output inverted 180°
Non-inverting amplifier gain	$A_{CL} = 1 + R_f / R_1$	Output in phase
Summing amplifier output	$V_o = -R_f(V_1/R_1 + V_2/R_2 + V_3/R_3)$	Adder
RC phase-shift frequency	$f = 1 / (2 \pi R C \sqrt{6})$	$\text{beta} = 1/29$, gain > 29
Wien bridge frequency	$f_0 = 1 / (2 \pi R C)$	Feedback = 1/3, $R_3/R_4 = 2$

10.2 Key Definitions at a Glance

Term	Definition
Feedback	The process of feeding a fraction of the output back to the input.
Positive feedback	Output fed back in phase, increasing the net input; used in oscillators.
Negative feedback	Output fed back in opposite phase, reducing the net input; used in amplifiers.
Barkhausen criterion	The condition $\beta A = 1$ required for self-sustained oscillation.
Oscillator	A circuit that produces an AC output of the desired frequency and waveshape without an external input.
Op-amp	A high-gain voltage amplifier with two inputs and one output that amplifies the difference of the inputs.
Virtual ground	The inverting input node of an op-amp, which sits at nearly 0 V without being physically grounded.
Virtual short	The near-zero voltage difference between the two op-amp inputs in a closed-loop circuit.
Slew rate	The maximum rate of change of the output voltage per unit time (V/ μ s).
CMRR	Common-mode rejection ratio = differential gain / common-mode gain.
Sinusoidal oscillator	An oscillator whose output varies sinusoidally.
Wien bridge oscillator	An RC bridge oscillator giving low-distortion sine waves at $f_0 = 1/(2 \pi R C)$.